

# B03 Engineering Data Modeling

From report models to implementation-ready dimensional contracts

## Fast-track purpose

Course 1 already taught star schemas, relationships, Date tables and model validation. B03 does not reteach those as Power BI topics. It converts them into engineering contracts: process/grain, key integrity, multiple facts/conformed dimensions and reproducible handoff.

## Locked learning loop

Why -> picture -> new words -> watch -> analyst-to-engineer transition -> follow once -> observe -> explain -> mistakes -> recover -> fresh task -> validate -> teach back -> protected review -> changed retry -> evidence/route.

INTERNAL QA RC1 • not course FINAL

## Contents & Study Route

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### Exact module route

For each lesson: book -> follow once -> matching practice sheet -> save genuine attempt/evidence -> protected review -> close review -> changed retry. After all four lessons: FreshRoute independent Mini-Lab -> Gate A. Gate B/C stay sealed unless explicitly assigned.

### Course 3 boundary

B03 prepares model contracts. Do not pre-learn dbt, surrogate-key implementation, SCD mechanics or CI here. C3-02 teaches those engineering implementations after C3 foundations/ingestion.

# B03-L01 — Analyst semantic model to engineering dimensional contract

## 1. Why it matters

A Power BI model can look correct while its physical grain contract is vague. An engineer must make the business process and one-row meaning explicit enough that a warehouse table can be built and tested from it.

## 2. What you are learning

Turn familiar star-schema thinking into a business-process + grain contract that controls facts, dimensions and future implementation.

## 3. Prerequisite

Course 1 PBM3: you can identify fact/dimension tables, state grain and recognize a star schema. You do not need warehouse implementation knowledge yet.

## 4. Picture it in your head

### Mental model

Imagine three boxes arriving at a warehouse door: one shipment box, one shipment-event box and one monthly-target box. They cannot be poured into one container just because they share customer or region labels. Each box has a different "one row means..." sentence.

## 5. New words before we use them

**business process** — the real activity being measured, such as shipping a parcel

**grain** — the exact real-world meaning of one fact row

**fact table** — rows of measurement/events at one consistent grain

**dimension** — descriptive context used to group/filter facts

**dimensional contract** — the written process, grain, keys, roles and rules that implementation must preserve

## 6. Watch / visual source

**Primary visual:** Chandoo, *Learn Data Modelling & Star Schema for Power BI in 20 minutes*, **02:18–09:39**. Use it only as a visual refresher. **After watching:** close the video and write one sentence each for Shipment, Shipment Event and Monthly Target grain.

[Chandoo video](#) | [Kimball Grain](#)

## Bridge transition

### Analyst -> Engineer

**What you already know as an analyst:** facts hold measurements and dimensions describe them.

**What changes in engineering:** the model begins with a business process and binding grain sentence before table roles.

**Why engineers care:** ETL/ELT, keys, tests and downstream totals all depend on that grain.

**Small worked example:** shipments = one row per shipment; shipment\_events = one row per shipment event; monthly\_targets = one row per month-region.

**Real engineering version:** C3-02 will physically implement these contracts in warehouse/dbt models; B03 stops at a correct implementation-ready design.

## 7. Follow with me once

Open Parcel Delivery source\_contracts.json and Source\_Inventory. For shipments, events and targets, write process + one-row grain first. Only then choose Fact/Dimension/Bridge roles.

## 8. What you should see

Three separate fact candidates remain separate. Customers and Services can describe shipment facts. No measure is placed into a table at an incompatible grain.

## B03-L01 — Practice to mastery

### 9. Why it worked

Declaring grain first prevents “table shape” from making the decision. Facts at different grains can still be analyzed together later through compatible conformed dimensions.

### 10. Common beginner mistakes

- Calling every large table a fact without stating its process/grain.
- Mixing shipment revenue with event rows, which repeats revenue.
- Starting from a diagram or Power BI relationship rather than the source process.
- Treating a monthly target as if it were one target per shipment.

### 11. If stuck

#### Recovery ladder

Read one file at a time. Say aloud: “one row represents \_\_\_\_.” If you cannot complete that sentence, do not choose the table role yet.

### 12. Small version alone

Invent a support operation with tickets, ticket status events and monthly team targets. Write three grain sentences and classify facts/dimensions.

### 13. Validate

Every fact has exactly one grain sentence. Every proposed measure is valid at that grain. Different grains are in separate facts.

### 14. Teach it back

Explain in 60–90 seconds why grain is a design contract rather than a Power BI setting.

### 15. Protected review + changed retry

#### Attempt protection

After saving your attempt, open Review B03-L01. Close it, then solve the changed support-domain task without copying wording.

### 16. Evidence / mastery + next route

#### Ready when

Saved process/grain table + role classification + one rejected mixed-grain design + changed retry + own-words explanation. PASS -> B03-L02; weak -> targeted grain repair only.

## B03-L02 — Key integrity, relationship paths and many-to-many bridges

### 1. Why it matters

In analytics, a relationship error may show up as a strange visual. In engineering, a wrong key/cardinality assumption can multiply rows, lose records, or silently corrupt every downstream metric.

### 2. What you are learning

Write explicit key/cardinality contracts, distinguish duplicate-key defects from unmatched foreign keys, and use bridge tables only for real membership relationships.

### 3. Prerequisite

B03-L01 mastered. From Course 1 PBM4 you already know 1:\* relationships and filter direction.

### 4. Picture it in your head

#### Mental model

Think of a train station. A dimension key is a named platform that should exist once. Fact rows are passengers with tickets pointing to that platform. A bridge is a legitimate transfer station for many-to-many membership—not a broom used to sweep duplicate mistakes away.

### 5. New words before we use them

**business key** — source-world identifier such as `customer_id`

**foreign key** — fact-side value that points to a dimension key

**cardinality** — how many rows on one side can match rows on the other

**fan-out** — unexpected row multiplication after a join

**bridge table** — a separate membership table connecting two many-valued entities at its own grain

### 6. Watch / visual source

**Primary visual:** Chandoo 11:50–18:07 for relationship/cardinality/filter flow. **After watching:** write the Customers→Shipments and Services→Shipments key contract before opening Power BI.

[Chandoo relationship segment](#) | [Microsoft many-to-many guidance](#)

### Bridge transition

#### Analyst -> Engineer

**What you already know as an analyst:** dimension (1) filters fact (\*).

**What changes in engineering:** uniqueness, FK meaning, unmatched-key policy and expected row-count behavior become explicit contracts.

**Why engineers care:** fan-out and bad keys corrupt data before a dashboard can warn you.

**Small worked example:** Customer[customer\_id] must be unique; Shipment[customer\_id] may repeat. Duplicate dimension key blocks the 1-side contract; unmatched shipment key is separately quantified.

**Real engineering version:** C3 will add physical constraints/tests and surrogate/history mechanics.

### 7. Follow with me once

In Parcel Delivery, audit Customers→Shipments and Services→Shipments. Then classify `customer_tags` at grain `customer_id + tag` as a legitimate membership bridge.

### 8. What you should see

You do not use DISTINCT to “fix” semantic duplication. You can explain why joining Events to Shipments can multiply shipment revenue.

## B03-L02 — Practice to mastery

### 9. Why it worked

The key contract separates two different problems: duplicate values on the supposed one-side versus foreign keys with no match. A real bridge represents a real many-to-many business relationship at its own grain.

### 10. Common beginner mistakes

- Switching to many-to-many because the one-side key is dirty.
- Using DISTINCT after a join without proving which duplicate is semantically wrong.
- Joining two fact tables directly because they share an ID.
- Using bidirectional filtering as a generic repair.

### 11. If stuck

#### Recovery ladder

Before drawing any relationship, write four lines: one-side table/key, uniqueness check, many-side FK, unmatched-key policy. If any line is unknown, investigate source grain.

### 12. Small version alone

Model Employee↔Skill membership with Employee, Skill and EmployeeSkill bridge. State each grain and composite bridge key.

### 13. Validate

One-side keys are unique/nonblank where required; bridge grain is real business membership; expected join row-count behavior is documented.

### 14. Teach it back

Explain bridge table vs many-to-many shortcut vs DISTINCT cleanup in 60–90 seconds.

### 15. Protected review + changed retry

#### Attempt protection

Save attempt -> Review B03-L02 -> close review -> changed Employee/Skill bridge task.

### 16. Evidence / mastery + next route

#### Ready when

Key/cardinality contract + fan-out risk note + bridge design + changed retry + explanation. PASS -> B03-L03.

## B03-L03 — Multiple grains, date roles and conformed dimensions

### 1. Why it matters

A warehouse often contains several business processes at different grains. They must remain separately correct yet still answer shared questions such as “by date, city or customer.”

### 2. What you are learning

Keep multiple facts independent, reuse conformed dimensions deliberately, and make role-playing dates explicit without mixing grains.

### 3. Prerequisite

B03-L01 and B03-L02 mastered. From C1 PBM5 you know a Date table; this lesson generalizes it across multiple facts.

### 4. Picture it in your head

#### Mental model

Picture three transparent maps laid over the same city: deliveries, delivery events and weekly targets. Each map has different dots, but they can align on shared street/date/city reference layers. Those shared layers are conformed dimensions.

### 5. New words before we use them

**conformed dimension** — a shared dimension definition reused consistently across more than one fact

**role-playing dimension** — one dimension used in different business roles, such as Order Date vs Delivery Date

**higher-grain fact** — a fact summarized at a broader level, such as week-city target

**drill across** — aligning separate fact results through shared dimensions rather than joining facts row-by-row

### 6. Watch / visual source

**Recommended visual:** Guy in a Cube, *Handling Relationships with Multiple Fact Tables in Power BI*, **00:00–02:29** and **06:04–08:33**. Skip the middle implementation alternatives for now. **After watching:** sketch separate facts sharing Date/City dimensions.

[Guy in a Cube](#) | [Kimball conformed dimensions](#) | [Microsoft star schema guidance](#)

### Bridge transition

#### Analyst -> Engineer

**What you already know as an analyst:** one Date table can support reporting and different relationships.

**What changes in engineering:** separate fact processes keep their own grain while shared dimension definitions become integration contracts.

**Why engineers care:** direct fact-to-fact joins create ambiguity and fan-out; conformance lets facts align safely.

**Small worked example:** Shipment fact uses Booking/Pickup/Delivery date roles; Event fact uses Event Date; Target fact uses Month/Region. Region/Date can be shared only with compatible definitions.

**Real engineering version:** C3-02 will implement dimensional tables, keys/history and serving models.

### 7. Follow with me once

Using Parcel Delivery, draw three independent fact boxes (Shipment, Event, Target). Around them place only dimensions that legitimately describe each fact. Label Booking/Pickup/Delivery date roles.

### 8. What you should see

No direct fact-to-fact relationship is required. Targets stay month-region grain. The Date definition is shared but each business role is named.

## B03-L03 — Practice to mastery

### 9. Why it worked

Conformed dimensions provide common descriptive coordinates. Keeping facts independent preserves their measurements while still allowing aligned analysis at compatible dimensions.

### 10. Common beginner mistakes

- Joining facts directly because they share shipment/customer/date columns.
- Expanding higher-grain targets down to transaction rows.
- Using one ambiguous Date relationship for three business dates.
- Calling two similarly named dimensions “conformed” when definitions/domains differ.

### 11. If stuck

#### Recovery ladder

Query each fact alone first. Ask: which dimensions are single-valued for one row of this fact? Only then decide what can be shared.

### 12. Small version alone

Design Orders, Web Events and Weekly Sales Targets sharing Customer/Date/Region only where valid. Name every fact grain and date role.

### 13. Validate

Each fact is independently queryable; shared dimensions have compatible definitions; date roles are explicit; no mixed-grain measure is required.

### 14. Teach it back

Explain “separate facts + conformed dimensions” to a junior analyst in 60–90 seconds.

### 15. Protected review + changed retry

#### Attempt protection

Save attempt -> Review B03-L03 -> close -> changed three-fact e-commerce task.

### 16. Evidence / mastery + next route

#### Ready when

Multi-fact diagram + conformed-dimension list + date-role map + rejected direct-fact join + changed retry. PASS -> B03-L04.

## B03-L04 — Model reconciliation and engineer handoff

### 1. Why it matters

A model is not ready for engineering because the diagram is pretty. Another person needs enough evidence to reproduce the design, test it against raw controls and know which assumptions are still risky.

### 2. What you are learning

Build an implementation-ready handoff packet with raw-to-model reconciliation, quality policies, ownership, rejected alternatives and explicit C3 boundaries.

### 3. Prerequisite

B03-L01-L03 mastered. From C1 PBM6/PBM7 you already know model validation and refresh hygiene.

### 4. Picture it in your head

#### Mental model

Imagine handing construction drawings to a builder. A picture of the finished house is not enough. The builder needs dimensions, materials, load assumptions, checks and what to do when a measurement does not match.

### 5. New words before we use them

**reconciliation** — proving important source counts/totals agree with modeled outputs under declared rules

**control total** — known count/sum/check used to detect silent loss or multiplication

**assumption owner** — person/team responsible for confirming a business rule

**handoff contract** — the design/evidence another engineer can implement without guessing

**rejected alternative** — a plausible design you deliberately did not choose, with the failure/trade-off documented

### 6. Watch / visual source

**Optional visual:** Chandoo 20:19–21:32 on why proper modeling matters. **After watching:** write one failure that a polished dashboard cannot repair. The lesson/practice packet remains the primary teacher for reconciliation/handoff.

[Chandoo closing segment](#) | [Kimball dimensional techniques](#)

### Bridge transition

#### Analyst -> Engineer

**What you already know as an analyst:** validate model row counts, keys, totals and refresh behavior.

**What changes in engineering:** those checks become a reproducible handoff contract with assumptions, ownership and failure policy.

**Why engineers care:** downstream implementation cannot rely on "it looked right in my PBIX."

**Small worked example:** before trusting Parcel Delivery, reconcile 72 shipments, 216 events, 12 targets, 28 customer-tag memberships and shipment revenue 26,571.6.

**Real engineering version:** C3 will convert these controls into automated tests/CI and physical model implementation.

### 7. Follow with me once

Complete Engineering\_Handoff and Reconciliation in the practice workbook. Record source controls before any relationship-dependent total. Reject one tempting design and name the exact failure.

### 8. What you should see

Another learner can state every fact grain, key contract, bridge/date roles, raw controls and open assumption without seeing your Power BI file.

## B03-L04 — Practice to mastery

### 9. Why it worked

Independent controls catch loss/multiplication that visuals can hide. Explicit assumptions separate business uncertainty from technical correctness and make change safer.

### 10. Common beginner mistakes

- Using one matching revenue total as proof of the whole model.
- Leaving unmatched keys or data-quality exceptions undocumented.
- Writing “best practice” without naming the failure/trade-off it avoids.
- Handing over screenshots without process/grain/key/control definitions.

### 11. If stuck

#### Recovery ladder

Start the handoff as a checklist: process -> grains -> keys -> dimensions -> bridge -> data roles -> relationships -> controls -> assumptions -> rejected alternative. Fill unknowns honestly.

### 12. Small version alone

Audit a fresh model sketch that lacks grain/key/control details. Write what the implementer would need before coding.

### 13. Validate

Raw controls match; fact/dimension/bridge design is reproducible; unresolved quality issues and assumption owners remain visible; no hidden C3 prerequisite is inserted.

### 14. Teach it back

Give a 2-minute engineering handoff defense without the lesson/review open.

### 15. Protected review + changed retry

#### Attempt protection

Save attempt -> Review B03-L04 -> close -> changed handoff-audit task.

### 16. Evidence / mastery + next route

#### Ready when

Completed handoff + reconciliation + assumption owner + rejected alternative + 2-minute defense + changed retry. PASS -> FreshRoute Mini-Lab.

## B03 Quick Reference & Assessment Route

| Question         | Engineering answer                                                       |
|------------------|--------------------------------------------------------------------------|
| Business process | Which real activity is being measured?                                   |
| Grain            | What exactly does one fact row mean?                                     |
| Key contract     | Which key is unique on the dimension side; which FK appears in facts?    |
| Bridge           | Does this table represent real many-to-many membership at its own grain? |
| Multiple facts   | Do different processes/grains stay in separate fact tables?              |
| Conformance      | Which dimension definitions are reused consistently across facts?        |
| Date roles       | Which business date does each fact key represent?                        |
| Reconciliation   | Which independent counts/totals prove no loss or multiplication?         |
| Handoff          | Could another engineer implement the model without guessing?             |

### Fresh Mini-Lab

Open B03\_FRESHROUTE\_MINILAB\_DATA\_RC1.zip and B03\_FRESHROUTE\_DIMENSIONAL\_MODEL\_MINILAB\_RC1.xlsx. Solve independently. The dataset is different from Parcel Delivery and from all Gate domains.

### Protected Gate route

Pass all four lessons + FreshRoute Mini-Lab -> Gate A Fleet Maintenance. If Gate A is weak: save attempt -> open Review A -> targeted repair. Gate B opens only by explicit full-retest assignment; same protection applies to Gate C. Completion percentage cannot bypass the Gate.